



Transforming banking: Examining the role of AI technology innovation in boosting banks financial performance

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ABSTRACT

With the ongoing advancement of technology, artificial intelligence is increasingly being integrated into banking and finance, with the transformative potential to reshape the financial performance of economies worldwide. This research investigates the dynamic relationship between banking and finance artificial intelligence technology innovation and banks' financial performance across 20 countries using the feasible generalized least squares and the generalized method of moments techniques. The results show that banking and finance artificial intelligence technology innovation positively impacts banks' return on assets, highlighting its role in enhancing financial performance. The interaction term between artificial intelligence innovation and economic growth emphasizes their collaborative positive impact on financial performance. Mediation analysis highlights information and communication technology development's role in transforming artificial intelligence innovation into improved financial outcomes. Considering lagged effects, initial innovation surges correlate with improved financial performance, but prolonged exposure leads to diminishing returns. Moreover, the findings indicate that non-performing loans negatively affect financial performance, underscoring the importance of asset quality. Additionally, regulatory capital and economic growth are positively associated with financial performance, while government regulations exhibit a negative impact. This study highlights the essential role of artificial intelligence technology innovation in banking and finance, emphasizing the need to consider economic and technological factors for maximizing its benefits in enhancing financial performance. Policy recommendations include promoting an artificial intelligence innovation ecosystem, adapting regulatory frameworks, and investing in information and communication technology infrastructure to harness artificial intelligence innovation's benefits while addressing associated challenges.

1. Introduction

In the ever-evolving landscape of banking and finance, the integration of artificial intelligence (AI) technology has emerged as a game-changer, revolutionizing traditional practices and reshaping industry dynamics (Subburayan, Sankarkumar, Singh, & Mushi, 2024). As financial institutions across the globe navigate the digital transformation journey, the role of AI innovation in enhancing banks' financial performance has become a focal point of research and scrutiny

(Shiyyab, Alzoubi, Obidat, & Alshurafat, 2023). The finance industry is on the verge of a technological transformation, with AI set to revolutionize traditional banking methods and propel unparalleled efficiency improvements and client-focused innovations. The proliferation of AI-driven applications, ranging from predictive analytics and risk management to customer service and fraud detection, has ushered in a new era of opportunity and competition for banks worldwide (Eni, Chaudhary, Raparathi, & Reddy, 2023). As banks grapple with evolving customer expectations, regulatory demands, competitive pressures, and

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the strategic adoption of AI technology has emerged as a pivotal driver of innovation and performance enhancement (Truby, Brown, & Dahdal, 2020). However, amidst the fervent enthusiasm surrounding AI's transformative potential, a critical question looms large: What is the true impact of banking and finance AI technology innovation on banks' financial performance? In an increasingly competitive marketplace, banks must leverage AI technology to drive operational efficiency, enhance customer experience, and seize new revenue opportunities (Dutta, Pramanik, Datta, & Kirtania, 2022). Regulatory authorities are grappling with the challenges posed by AI-driven innovation, including data privacy, algorithmic bias, and systemic risks (Kreps & Rao, 2023).

Understanding the impact of AI on financial performance is crucial for informing regulatory responses and ensuring a balance between innovation and risk mitigation (Sullivan & Wamba, 2024). Financial institutions are making significant investments in AI technology to gain a competitive edge and adapt to changing market dynamics (Zhao, Tsai, & Wang, 2019). Assessing the fiscal ramifications of AI advancement is crucial for steering investment choices and enhancing resource distribution. The policy relevance of this topic is unequivocal, resonating with the imperatives of the digital age and the imperatives of the financial industry. In an era marked by unprecedented technological disruption and rapid digital transformation (Zaki, 2019), policymakers and regulators face a daunting challenge: how to navigate the evolving landscape of AI innovation while safeguarding consumer interests, ensuring financial stability, and promoting inclusive growth. At its core, this study seeks to empower policymakers, regulators, financial institutions, and researchers with the knowledge and evidence base needed to harness the full potential of AI for driving sustainable growth, enhancing financial stability, and fostering innovation in banking.

This study aims to investigate the impact of AI technology innovation on banks' financial performance across 20 diverse countries, addressing three key questions: how banking and finance AI technology innovation influences banks' financial performance in different economic, regulatory, and technological contexts; the mediating effects of information and communication technology (ICT) development and the moderating impacts of economic growth on this relationship; and the evolving effects of AI technology innovation over time. Employing a longitudinal panel data analysis, the research explores both immediate and lagged impacts of AI innovation, offering valuable insights for strategic planning. This study highlights the need for an empirical assessment of AI's transformative potential in banking, considering the diverse landscapes of the selected countries. This research is essential for guiding policymakers, regulators, and financial institutions on leveraging AI for sustainable growth, financial stability, and innovation in the banking sector. Within the framework of the 20 chosen countries, variables such as AI technology innovation, banks' financial performance, ICT development, economic growth, and regulatory environment are investigated. Understanding the global dynamics of AI integration in banking requires an awareness of these elements. Through applications like fraud detection and predictive analytics, AI technology innovation improves customer service, lowers costs, and increases efficiency. Nations like the United States, China, and Japan lead in AI research and development due to their significant investments and advanced technological infrastructures. This study offers a thorough analysis with novel insights into AI's impact on banking across all stages of economic growth by examining diverse economic, regulatory, and technological contexts. Additionally, the selection of countries with varying regulatory frameworks and cultural diversity enhances the generalizability of the findings, providing a comprehensive understanding of AI integration in global financial services.

This study embarks on a pioneering journey to unravel the complexities of the relationship between banking and finance AI technology innovation and banks' financial performance, offering a fresh perspective that sets it apart from existing literature. By venturing into uncharted territory, this study makes significant contributions to empirical research by advancing the understanding of the impact of AI technology

innovation on banks' financial performance across diverse global contexts. First, it addresses the gap in existing literature, which has primarily focused on single-country analyses or firm-level studies, by employing a comprehensive panel data analysis of 20 countries. This approach provides a broader perspective on the global dynamics of AI innovation in the banking sector. Second, the research delves into the mechanisms through which AI influences financial performance, considering mediating factors like ICT development and moderating variables such as economic growth and regulatory environments. This novel analysis uncovers the complex pathways through which AI technology translates into tangible financial outcomes. Third, the study adopts a longitudinal perspective, examining the lagged effects of AI innovation over time, thereby capturing the evolving nature of its impact on financial performance. This comprehensive approach not only enhances the robustness of the findings but also offers valuable insights for long-term strategic planning and policymaking in the banking industry, highlighting the transformative potential of AI technology in driving sustainable growth and competitiveness. The findings indicate that AI technology innovation in banking and finance results in an increase in banks' financial performance. Additionally, economic growth interacts with AI technology innovation, further enhancing banks' financial performance. Moreover, ICT development serves as a mediator between AI technology innovation in banking and finance and banks' financial performance. The impact of AI technology innovation on banks' financial performance also varies across different lag periods. These findings highlight that integrating AI into banking and finance not only enhances financial performance but also requires a supportive economic and technological infrastructure to achieve its full potential.

Furthermore, the major findings of this study are as follows: 1. Banking and finance artificial intelligence technology innovation positively impacts banks' return on assets. 2. The interactive term between artificial intelligence innovation and economic growth emphasizes their collaborative positive impact on financial performance. 3. Non-performing loans negatively affect financial performance, underscoring the importance of asset quality. 4. Regulatory capital and economic growth are positively associated with financial performance, while government regulations exhibit a negative impact.

The subsequent sections of this study are organized as follows: [Section 2](#) presents the theoretical background, literature review, and research hypotheses. [Section 3](#) details the methodology, model, and data. [Section 4](#) outlines the empirical results and their interpretation. Finally, [Section 5](#) offers concluding remarks and policy recommendations.

2. Theoretical background, literature review and research hypotheses

This section outlines the theoretical channels and research hypotheses through which AI technology innovation in banking and finance can impact the financial performance of banks. Additionally, it offers a synopsis of empirical research exploring the relationship between AI technology innovation and banks' financial performance.

2.1. Theoretical frameworks

In the dynamic landscape of global banking, understanding how AI technology innovations reshape financial performance requires a multifaceted approach: from leveraging unique AI capabilities as strategic resources (resource-based view), to tracing their diffusion patterns across markets (diffusion of innovation theory), and navigating the regulatory, ethical, and cultural landscapes shaping adoption (institutional theory). The resource-based view (RBV) offers a theoretical lens to understand how banking and finance artificial intelligence technology innovation can influence banks' financial performance. According to RBV, firms gain competitive advantage and superior performance by leveraging unique and valuable resources, such as AI capabilities, that

are difficult for competitors to imitate or substitute (Barney, 1991). In the context of this study, AI technology innovation represents a strategic resource that can enhance banks' ability to streamline operations, improve risk management, and personalize customer experiences, thereby contributing to superior financial performance. RBV integrates seamlessly with the study's focus on AI technology innovation in banking. By viewing AI as a strategic resource, RBV aligns with the argument that effective deployment of AI can lead to competitive advantages for banks, thereby improving financial performance (Rahman, 2023). The study can analyze how different AI capabilities (e.g., machine learning for credit risk assessment, and chatbots for customer service) contribute to enhancing specific aspects of financial performance metrics like return on assets (ROA) and return on equity (ROE). RBV may conflict if the study finds that the adoption of AI technologies by banks does not result in sustained competitive advantage or superior financial performance. This conflict could surface if the implementation of AI is not aligned with strategic objectives or if competitors quickly replicate AI innovations, diminishing their strategic value over time.

Diffusion of innovation theory (Dol) provides a theoretical framework for examining the adoption and diffusion of AI technology innovation within the banking sector and its subsequent impact on financial performance (Rogers, Singhal, & Quinlan, 2014). Dol posits that innovation adoption follows a predictable pattern, starting with early adopters and gradually spreading to the mainstream market. By applying Dol, this study can analyze the factors influencing the adoption of AI innovation by banks across different countries, the rate of diffusion over time, and its implications for financial performance. Dol integrates with the study by framing AI technology adoption as a process influenced by factors such as technological complexity, organizational readiness, and market demand. Understanding these adoption dynamics is crucial for assessing how quickly AI innovations can contribute to improving financial performance metrics like profitability and efficiency ratios (Shiyyab et al., 2023). Dol may conflict if the study finds that AI adoption rates are inconsistent or if certain banks experience barriers to adopting AI technologies despite their potential benefits. Conflicting results could occur if cultural or regulatory factors slow down the diffusion of AI innovations, limiting their impact on financial performance outcomes.

Institutional theory offers a theoretical framework to explore how regulatory, cultural, and institutional factors influence the adoption and impact of AI technology innovation in the banking sector across different countries (Scott, 1995). Institutional pressures, such as regulatory mandates, industry norms, and societal expectations, shape banks' decisions regarding AI adoption and their subsequent financial performance (Engvall, 2024). By applying Institutional theory, this study can analyze how institutional factors interact with AI innovation to influence banks' financial performance in diverse national contexts. Institutional theory integrates with the study by highlighting the role of regulatory frameworks and industry norms in shaping AI adoption strategies and practices in banking. By understanding these institutional pressures, the study can analyze how compliance with regulations and alignment with industry standards influence the effectiveness of AI implementations on financial metrics. Institutional theory may conflict if the study finds that regulatory barriers or cultural resistance impede banks' ability to adopt AI technologies effectively. Conflicting outcomes could arise if varying regulatory environments across countries create uneven playing fields for AI adoption or if societal expectations discourage banks from fully leveraging AI's potential benefits.

2.2. Literature review

Several studies have examined various factors contributing to the financial performance of banks. These factors include non-performing loans in banks (Takahashi & Vasconcelos, 2024), regulatory capital in banks (De Bandt, Camara, Maitre, & Pessarossi, 2018), economic growth (Alam, Rabbani, Tausif, & Abey, 2021), government regulations (Bătae,

Dragomir, & Feleagă, 2021), and ICT development (Al-Busaidi & Al-Muharrami, 2021). These studies provide a rich context for understanding how banking and finance AI technology innovation (BFAITI) might influence financial outcomes. Takahashi and Vasconcelos (2024) identified non-performing loans (NPLs) as a critical determinant of banks' financial performance. They demonstrated that NPLs negatively affect return on assets (ROA), thereby reducing overall profitability. This finding was echoed by (Duong, Tran, Nguyen, & Pham, 2023), who found that NPLs diminish ROA by 0.075 across 37 Vietnamese banks from 2005 to 2020 using fixed effect and two-step system generalized method of moments models. Regulatory capital's impact on bank profitability is complex. Tran, Lin, and Nguyen (2016) revealed that regulatory capital exerts a dual effect: it negatively impacts banks with lower capital but positively influences those with higher capital. Their study, spanning U.S. banks from 1996 to 2013, utilized quarterly panel data to uncover these dynamics. Economic growth has been shown to positively correlate with bank profitability. Ledhem and Mekidiche (2020), employing the generalized method of moments, identified a positive relationship between economic growth and return on equity (ROE) in Islamic banks across Malaysia, Indonesia, Brunei, Turkey, and Saudi Arabia from 2014 to 2018. Regulatory frameworks can boost or pose challenges to bank performance. (Nasreen, Gulzar, Afzal, & Farooq, 2024) found a negative correlation between government regulations, corruption, and bank performance in an analysis of 309 banks across 15 Asian countries from 2010 to 2020. This study highlights the complexities regulatory environments introduce to banking operations. ICT development's role in bank performance is complex and context-dependent. Aguegboh, Agu, and Nnetu-Okolieuwa (2023) discovered that while ICT adoption negatively correlates with ROA, it positively impacts return on earnings and net interest margin across 35 sub-Saharan African countries. This novel finding highlights ICT's multifaceted impact on bank performance.

With the increasing integration of artificial intelligence in banking, recent studies have focused on its socioeconomic, economic, and financial impacts. Shiyyab et al. (2023) analyzed 15 Jordanian banks listed on the Amman stock exchange from 2014 to 2021 and found that AI significantly enhances financial performance, boosting both ROA and ROE. Similarly, Königstorfer and Thalmann (2020) reported that AI reduces losses in lending, enhances payment security, automates compliance tasks, and refines customer targeting, all of which bolster overall performance. Ris, Stankovic, and Avramovic (2020) showed that integrating AI technologies, such as virtual assistants and machine learning algorithms, improves bank profitability and performance while reducing dependence on human labor. These technologies also aid in fraud prevention and facilitate more informed decision-making processes. This demonstrates the AI-driven operational enhancements in banks' performance. (Umamaheswari & Valarmathi, 2023) emphasized the role of AI-powered virtual assistants in enhancing profitability, improving performance, and reducing human resource dependency, particularly in the banking sector. This shows AI's virtual assistants and performance in the banking sector. (Chhaidar, Abdelhedi, & Abdelkafi, 2023) highlighted the positive impact of fintech innovations, including the Internet of Things, big data, and AI, on bank profitability. Their study of 23 European banks from 2010 to 2019, using the fully modified ordinary least squares model, found a significant correlation between these technologies and enhanced financial performance. According to Abrokwhah-Larbi and Awuku-Larbi (2024), artificial intelligence in marketing markedly boosts financial outcomes, customer satisfaction, internal operational efficiency, and developmental progress by leveraging artificial intelligence in marketing components such as the Internet of Things (IoT), collaborative decision-making platforms (CDMP), virtual and augmented reality (VAR), and personalization techniques. Their research is grounded in an analysis of 225 small and medium enterprises (SMEs) in Ghana, utilizing the resource-based view framework. Madhura, Panakaje, Kambali, and Parvin (2024) asserted that AI technologies hold immense promise for reducing expenses across

numerous banks, thereby boosting financial performance. Additionally, they help avert fraudulent transactions, enhance anti-money laundering efforts, improve client identification and verification processes, deliver tailored insights and recommendations, strengthen customer relationships via chatbots and voice assistants, and emulate human interactions through AI technologies.

Despite the growing prominence of artificial intelligence technology in banking and finance, empirical research specifically examining its direct impact on banks' financial performance across multiple countries remains scarce. Existing studies often focus on single-country analyses or overlook the complex interactions between AI innovation, ICT development, economic growth, and regulatory environments in shaping financial outcomes. This study aims to fill this gap by investigating how AI technology innovation influences the financial performance of banks across a diverse panel dataset encompassing 20 countries by utilizing a novel variable for measuring banking AI technology innovation. By leveraging cross-country panel data analysis, the research seeks to provide empirical evidence on the pathways through which AI innovations enhance banks' operational efficiencies, risk management practices, and customer engagement strategies. This approach not only addresses the gap in empirical literature but also contributes to strategic decision-making within the banking sector by revealing the conditions under which AI adoption leads to measurable improvements in financial metrics like return on assets and return on equity. Through its exploration of mediating factors such as ICT development and moderating variables including economic growth, the study aims to uncover innovative pathways that translate AI innovation into tangible financial outcomes for banks, thereby advancing knowledge in both academic and practical spheres of banking and finance.

2.3. Research hypotheses

In the era of digital transformation in banking, the integration of AI technologies promises to revolutionize financial performance through enhanced operational efficiencies, strategic decision-making, and customer-centric innovations. How exactly does banking and finance AI technology innovation influence banks' financial performance across global markets, and what role do economic growth and ICT development play in shaping this relationship? Banking and finance AI technology

innovation (BFAITI) encompasses a range of applications, including automated decision-making, predictive analytics, and personalized customer services, which can streamline operations, optimize resource allocation, and mitigate risks within financial institutions (Polak, Nelischer, Guo, & Robertson, 2020). By leveraging AI-driven insights and automation, banks can enhance efficiency, reduce costs, and improve profitability, ultimately leading to improved financial performance (see Fig. 1).

Hypothesis 1. BFAITI positively influences banks' financial performance.

Economic growth creates a conducive environment for the adoption and implementation of AI technology innovation within the banking sector (Reis Mourão & Popescu, 2023). As economies expand, banks have greater financial resources, technological infrastructure, and market demand to invest in AI-driven solutions. Additionally, economic growth may stimulate innovation ecosystems, fostering collaboration between financial institutions, technology firms, and research institutions, thereby accelerating the development and diffusion of AI innovations in banking. Economic expansion facilitates the growth of BFAITI to improve banks' financial performance.

Hypothesis 2. Economic growth moderates the positive relationship between BFAITI and banks' financial performance.

BFAITI contributes to the development and adoption of ICT infrastructure within financial institutions (Rahman, Ming, Baigh, & Sarker, 2023). As banks invest in AI-driven technologies, they often need to upgrade their ICT infrastructure, such as data storage, processing capabilities, and connectivity, to support the implementation and integration of AI systems. AI innovations enhance ICT development enabling banks to harness the full potential of AI, leading to improvements in operational efficiency, decision-making processes, and customer experiences, ultimately translating into enhanced financial performance. In brief, BFAITI enhances banks' financial performance by advancing ICT development.

Hypothesis 3. ICT development mediates the relationship between BFAITI and banks' financial performance.

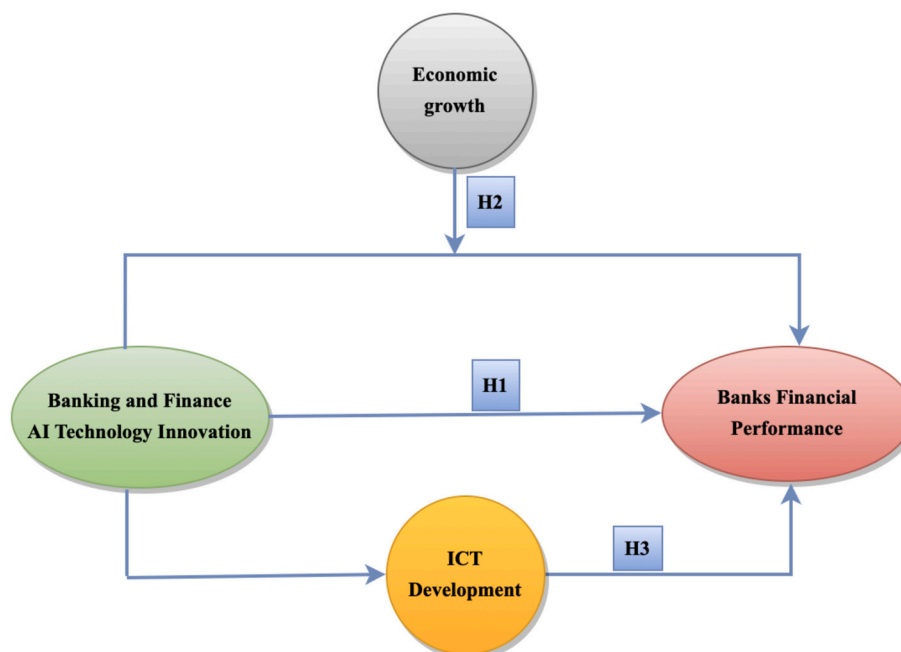


Fig. 1. Theoretical mechanisms.

3. Methodology, model, and data

3.1. Methodology

In this section, the investigation outlines the econometric techniques employed to scrutinize the influence of banking and finance AI innovations on the financial performance of banks. Moreover, it explores the moderating impact of economic expansion and the intermediary role of information and communication technology advancement, identified as notable factors affecting bank financial performance. The segment begins with preliminary examinations, which include assessments for slope heterogeneity (SH), heteroscedasticity, serial correlation, cross-sectional interdependence (CSD), and tests for stationarity. Subsequently, the analysis advances to assess the impact coefficients using the feasible generalized least squares methodology.

First, the study employs the CSD test based on the methodology outlined by Pesaran (2021) to identify any interdependence among cross-sectional units. In panel data analyses, assessing for CSD before selecting the optimal estimator has become a conventional practice. Researchers have employed CSD to determine the appropriate estimator in their panel analyses over the years (Appiah, Ashraf, Tiwari, Gyamfi, & Onifade, 2023). Panel studies involving countries with similar trends have a high likelihood of spillover effects, where events in one nation influence the economies of others due to growing economic interdependence (Gyamfi, Ozturk, Bein, & Bekun, 2021).

Next, the study utilizes the SH test developed by Pesaran and Yamagata (2008) to determine whether there are heterogeneous slope coefficients present in the panel data. The issue of slope heterogeneity, stemming from diverse economic and demographic structures, is also pivotal in panel data econometrics. Heterogeneity illustrates that parameters of interest vary across cross-sectional units. For instance, in this analysis, assuming uniform slope parameters in panel data comprising both extensive and limited panels is inappropriate. To prevent erroneous results from estimations, the assumption of homogeneity must be validated (Breitung & Das, 2005).

Subsequently, the study conducts a modified heteroscedasticity test following the approach proposed by Lütkepohl and Burda (1997) to check for the presence of heteroscedasticity, which refers to the non-constant variance of the error terms in the regression model. Heteroscedasticity can lead to inefficient estimates and affect the validity of statistical inferences (Papadopoulos, 2024). By applying this test, the study ensures the reliability and consistency of the estimated coefficients. Serial correlation tests based on the method outlined by Born and Breitung (2016), is used to explore any evidence of these phenomena within the dataset. This test is employed to detect serial correlation, which occurs when the residuals of a model are correlated across periods. Serial correlation can indicate model misspecification and lead to biased standard errors, resulting in misleading statistical inferences. Conducting this test helps to identify and address any autocorrelation in the dataset, ensuring robust and accurate model results.

Moving forward, the econometric methodology proceeds with a panel stationarity test using cross-sectional augmented Dickey-Fuller (CADF) and cross-sectional Im Pesaran & Shin (CIPS) tests, introduced by Pesaran (2007) which are second-generation panel stationarity tests specifically designed to address the issue of CSD. The results of the CSD assessment will dictate the type of examination to be conducted. If the variables are not interdependent across sections, the first-generation stationarity test is utilized. Conversely, if cross-sectional dependence exists, the second-generation stationarity test is warranted (Appiah, Karim, Naeem, & Lucey, 2022). The panel stationarity test is to determine whether the time series data used in the panel dataset are stationary, meaning their statistical properties such as mean and variance do not change over time (To, H. K and Chan, 2024). Stationarity is a crucial assumption in many econometric models, especially when dealing with panel data, to ensure valid and reliable inference.

Following the assessment of variable stationarity, we employed

feasible generalized least squares (FGLS) to analyze the impact elasticities of banking and finance AI technological innovation (BFAITI) on banks' financial performance. According to Bai, Choi, and Liao (2021), FGLS assumes that the residuals follow a first-order autoregressive process AR (1). By explicitly modeling the autoregressive process, FGLS effectively eliminates residual autocorrelation, resulting in more efficient estimates. This feature is crucial in panel data analysis where observations across time may exhibit temporal dependencies, ensuring that the estimates of BFAITI's effects on financial performance are not biased due to autocorrelation. Compared to ordinary least squares (OLS), FGLS provides more efficient and consistent parameter estimates, especially when the assumptions of homoscedasticity and independence of errors are violated (Ghaemi & Hosseini, 2023). This is particularly relevant in dynamic settings like banking where the effects of BFAITI may evolve. Moreover, FGLS is adaptable and capable of accommodating various forms of error structure, including heteroscedasticity, serial correlation, and CSD in panel data analysis (Imran et al., 2024). These capabilities ensure that the model accurately captures the diverse impacts of BFAITI on banks' financial performance across the studied economies while handling longitudinal study issues such as heteroscedasticity (varying levels of variance across observations) and serial correlation (correlation between error terms over time). In summary, FGLS is not only technically robust but also methodologically advanced, making it an ideal choice for this study to produce accurate, reliable, and policy-relevant insights into how BFAITI influences banks' financial performance across diverse economic contexts. According to Fomby et al. (1984), the generalized least squares framework adopts the structure

$$y = X\beta + e \quad (1)$$

Here, X is $T \times K$ matrix of non-stochastic elements of rank K satisfying the limit condition

$$\lim_{n \rightarrow \infty} \left(\frac{X' \Omega^{-1} X}{T} \right) = \Omega_{\Omega} \quad (2)$$

Ω_{Ω} being a $K \times K$ finite and nonsingular matrix. The error vector e satisfies the properties

$$Ee = 0 \quad (3)$$

and

$$Eee' = \sigma^2 \Omega \quad (4)$$

Suppose, for instance, that instead of being independently and identically distributed with a mean of zero and variance of σ^2 , the errors e , adhere to the first-order autoregressive process.

$$e_t = \rho e_{t-1} + u_t, |\rho| < 1, \quad (5)$$

with u_t independently and identically dispersed and have a zero average and finite variance σ^2 .

The research model employing linear FGLS can be represented as:

$$Y_{it} = \beta_0 + \beta_1 X_{it} + B_2 Z_{it} + \epsilon_{it} \quad (6)$$

In this study, Y_{it} denotes the dependent variable (banks' financial performance), X_{it} represents the independent variable (banking and finance AI technological innovation), Z_{it} indicates the control variables, and ϵ_{it} is the error term adjusted for potential autocorrelation using the AR (1) process.

3.2. Model

Apart from artificial intelligence, various other factors influence banks' financial performance, such as bank non-performing loans (Partovi & Matousek, 2019), bank regulatory capital to risk-weighted assets (Nsanyan Sandow, Duodu, & Oteng-Abayie, 2021), economic

growth (Alam, Rabbani, Tausif, & Abey, 2021), and government regulations (Shen & Chang, 2006). To thoroughly investigate the effects of banking and finance AI technological innovation on banks' financial performance, we incorporate these factors as control variables in our measurement model, which is outlined as follows:

$$BROA_{it} = \beta_0 + \delta_1 BFAITI_{it} + \gamma X_{it} + \varepsilon_{it} \quad (7)$$

In Eq. (7), the subscripts i and t denote specific countries and years, respectively. $BFAITI_{it}$ stands for banking and finance AI technological innovation, X_{it} control variables including banks non-performing loans (BNPL), bank regulatory capital (BRC), economic growth (GDPG) and regulations (REG). The symbol β_0 represents the constant term, while δ_1 and γ are the coefficients associated with each control variable to be estimated. Finally, ε_{it} represents the error term.

Furthermore, we explore whether the impact of banking and finance AI technological innovation on banks' financial performance is influenced by economic growth and verify Hypothesis 2. Additionally, we introduce an interaction term between banking and finance AI technological innovation and economic growth. As a result, the Eq. (7) is modified as follows:

$$BROA_{it} = \beta_0 + \delta_1 BFAITI_{it} + \gamma X_{it} + \psi(BFAITI_{it} * GDPG_{it}) + \varepsilon_{it} \quad (8)$$

In this equation, ψ denotes the coefficient of the moderating variable that requires estimation.

Furthermore, the paper delves into the mechanism by which BFAITI influences the financial performance of banks and validates Hypothesis 3. Drawing on the study by Li and Zhang (2023) on mediating effects, the subsequent equation for assessing the mediating effect is formulated:

$$MV_{it} = \beta_0 + \delta_2 BFAITI_{it} + \gamma X_{it} + \varepsilon_{it} \quad (9)$$

$$BROA_{it} = \beta_0 + \delta_3 BFAITI_{it} + \varphi MV_{it} + \gamma X_{it} + \varepsilon_{it} \quad (10)$$

Here, the term MV_{it} denotes the mediating variable, which represents ICT development (ICTD) and φ represent the coefficient of the mediating variable to be estimated.

To evaluate the lagged effects of banking and finance AI technology innovation on banks financial performance, this study integrates the strength of banking and finance AI technology innovation from the first to fifth lag periods as independent variables into Eq. (7), resulting in Eq. (11) as expressed below:

$$BROA_{it} = \beta_0 + \delta_1 BFAITI_{it-k} + \gamma X_{it} + \varepsilon_{it}$$

$$k = 1, 2, 3, 4, 5 \quad (11)$$

where $k = 1, 2, 3, 4$, and 5 are the lagged periods. The rest of the variables remain unchanged.

3.3. Data and variables declaration

From 2010 to 2020, this research investigates the connection among various factors, including bank return on assets (BROA), banking and finance artificial intelligence technology innovation (BFAITI), bank non-performing loans (BNPL), bank regulatory capital (BRC), economic growth (GDPG), and government regulations (REG) across 20 economies. Additionally, the study delves into the moderating impact of economic growth and the mediating role of ICT development (ICTD) on the relationship between banking and finance AI technological innovation (BFAITI) and banks' financial performance (BROA). The chosen sample and time frame were determined based on data availability for the independent variable. Data related to all variables were extracted from the Global Financial Development (GFD) and World Development Indicators (WDI) within the World Bank database, as well as the Center for Security and Emerging Technology (CSET) (Center for Security and Emerging Technology, 2023a; World Bank, 2024). Table 1 presents a description of the variables, including their measurement and source.

Table 1

Variables, measurement and source.

Variable	Symbol	Unit measurement	Source
Banks financial performance	BROA	Bank return on assets percentage, after tax	GFD
Banking and finance AI technology innovation	BFAITI	Patent applications granted - field: banking and finance	CSET
Bank non-performing loans	BNPL	Percentage	GFD
Bank regulatory capital	BRC	Percentage	GFD
Economic growth	GDPG	GDP growth (annual %)	WDI
Government regulation	REG	Regulatory quality: estimate	WDI
ICT Development	ICTD	Composite of mobile cellular subscriptions (per 100 people) and individuals using the internet (% of population)	WDI

All variables are transformed to their natural lag.

3.3.1. Dependent variable

Bank financial performance is measured by *bank return on assets (BROA)*, expressed as a percentage after tax. ROA is a key metric used to evaluate a bank's profitability by indicating how efficiently it utilizes its assets to generate profits. A higher ROA signifies better financial performance. This variable serves as the primary outcome measure to assess the impact of AI technology innovation on banks' financial success (Haibo & Manu, 2022). Previous studies have employed BROA as a measure of financial performance (Adu, Abedin, Saa, & Boateng, 2024). The yearly ROA of banks in the sampled countries is depicted in Fig. 2.

3.3.2. Independent variable

Banking and finance AI technology innovation (BFAITI) measured by patent applications granted in the field of banking and finance reflects the level of innovation within the banking and finance sector specifically related to AI technology. This variable represents the innovative efforts and investments made by banks in adopting AI technology to enhance their operations, products, and services (Kaya, Schildbach, AG, & Schneider, 2019). Fig. 3 shows the sampled countries annually granted patents in the banking and finance sector. Numerous studies have utilized different proxies to represent AI technology including patents and keywords for AI-related terms (Shiyyab et al., 2023). This study uses AI patents specifically granted in the banking and finance sector which makes it different and novel from previous studies on AI and banks' performance.

3.3.3. Control variables

Bank non-performing loans to gross loans (BNPL) measures the proportion of non-performing loans relative to total gross loans and helps control for the risk associated with asset quality in banks' financial performance analysis (Galariotis, Kosmidou, Kousenidis, Lazaridou, & Papapanagiotou, 2021). Zeqiraj, Gurdgiev, Sohag, and Hammoudeh (2024) used this variable in their studies on bank performance and economic uncertainty.

Bank regulatory capital to risk-weighted assets (BRC) represents regulatory capital adequacy which is essential for ensuring the stability and solvency of banks. This variable assesses the capital reserves held by banks to cover potential losses, thus controlling for the impact of capital strength on financial performance (Athanasoglou, Brissimis, & Delis, 2008). Several researchers have incorporated this variable in their studies relating to banking and finance (Siddika & Haron, 2020).

Economic growth (GDPG) measured by GDP growth reflects the overall macroeconomic environment in which banks operate. It serves as a control variable to account for external economic factors that may influence banks' financial performance (Hakimi, Hamdi, & Khemiri, 2023). This variable has been utilized in numerous research to measure economic growth (Asongu & Biekpe, 2018).

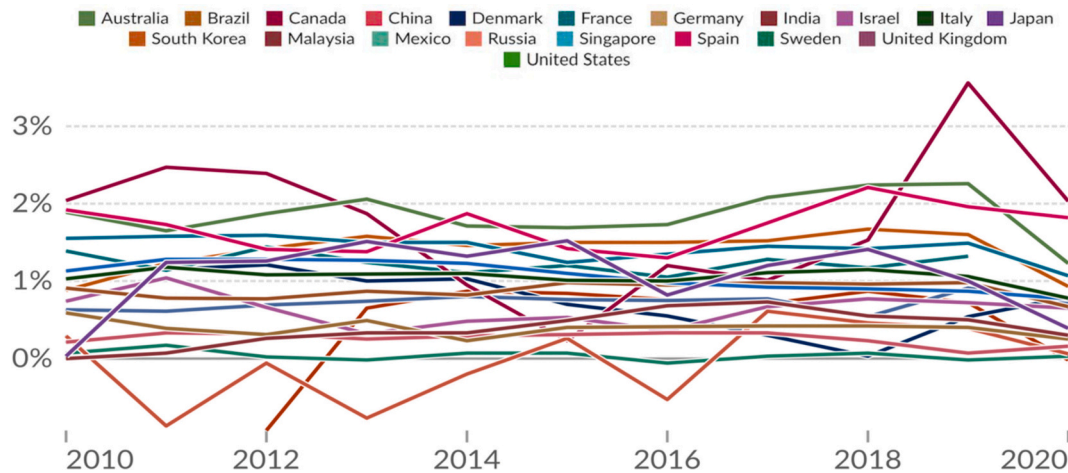


Fig. 2. Annual banks return on assets from 2010 to 2020. Source: (International Monetary Fund, 2023).

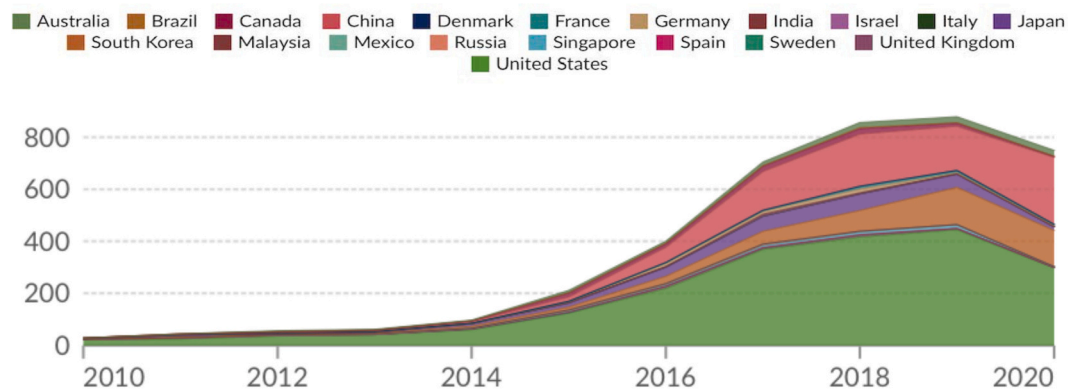


Fig. 3. Annual AI patents granted in the field of banking and finance from 2010 to 2020. Source: Center for Security and Emerging Technology (2023).

Government regulation (REG) measured by regulatory quality plays a crucial role in shaping the operating environment for banks. Regulatory quality serves as a control variable to capture the effect of regulatory frameworks on banks' financial performance (Anginer, Demircug-Kunt, & Zhu, 2014). According to Ferrari, Masetti, and Ren (2018), regulatory policies imposed by the government, including limitations on interest rates or constraints on lending procedures, can affect the interest earnings and net interest margins of banks.

3.3.4. Moderating variable

Economic growth (GDPG) measured by GDP growth can also moderate the relationship between AI technology innovation and banks' financial performance. Higher economic growth may amplify the positive impact of AI technology innovation on banks' profitability, while lower economic growth may dampen its effects (Banna & Alam, 2021).

3.3.5. Mediating variable

ICT Development (ICTD) measured by a composite of mobile cellular subscriptions and individuals using the internet acts as a mediating variable that explains the mechanism through which AI technology innovation influences banks' financial performance. Implementation of AI technologies enhances ICT infrastructure development, leading to improved operational efficiency and customer service quality, ultimately affecting banks' profitability (Shao, Lou, Wang, Mao, & Ye, 2022). The use of ICT variables including internet usage, mobile subscriptions, and fixed telephone subscriptions have been employed in several studies, especially banking and finance (Del Gaudio, Porzio, Sampagnaro, & Verdoliva, 2021).

4. Results and discussion

4.1. Preliminary tests

The summary statistics of the variables are provided in Table 2. Notably, the average for BROA is 0.65, with a standard deviation of 0.536. The range of BROA varies from a minimum of -1.526 to a maximum of 2.901 . In contrast, BFAITI has a relatively high mean of 18.586 , accompanied by a substantial standard deviation of 61.197 . Its range spans from a minimum of 0 to a maximum of 445 . Other variables such as BNPL, BRC, GDPG, REG, and ICTD exhibit mean values of 3.143 , 15.885 , 2.201 , 0.999 , and 191.418 , respectively. The correlation coefficients among these variables, as displayed in Table 2, are all below 0.5 , indicating the absence of multicollinearity concerns. Moreover, there is statistically significant evidence of a correlation between the dependent variable and all other factors, except for bank regulatory capital, at least at the 10% significance level based on the correlation coefficients. Lastly, BRAITI and GDPG show a positive correlation with BROA, while BNPL, BRC, and REG display a negative correlation with BROA, as depicted in Table 2.

The Pesaran and Yamagata (2008) SH, Lütkepohl and Burda (1997) heteroscedasticity, and Born and Breitung (2016) serial correlation tests are summarized in Table 3. These tests reveal statistical significance at the 1% level, indicating the presence of slope heterogeneity, heteroscedasticity, and serial correlation within the model.

Table 4 presents the results of the CSD, CIPS, and CADF tests. The results indicate that all variables demonstrate CSD within the panel data, signifying interdependence among countries in the panel data.

Table 2

Descriptive and correlation statistics.

Variable	BROA	BFAITI	BNL	BRC	GDPG	REG	ICTD
Mean	0.65	18.586	3.143	15.885	2.201	0.999	191.418
Std. Dev.	0.536	61.197	3.269	2.824	3.315	0.788	35.142
Minimum	−1.526	0	0.236	11.397	−11.167	−0.58	68.13
Maximum	2.901	445	18.064	26.884	14.52	2.252	248.584
Correlation statistics							
BROA	1						
BFAITI	0.141*	1					
BNL	−0.341***	−0.171*	1				
BRC	−0.0476	−0.0970	−0.259***	1			
GDPG	0.328***	0.0159	−0.135*	−0.229***	1		
REG	−0.247***	0.0160	−0.400***	0.364***	−0.156*	1	
ICTD	−0.138*	0.0242	−0.0370	0.343***	−0.353***	0.516***	1

Note: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.**Table 3**

SH, heteroscedasticity and serial correlation tests.

Test	Statistics	P-value
Slope Homogeneity	2.475**	0.013
Slope Homogeneity (Adjusted)	4.739***	0.001
Modified Wald test for groupwise heteroskedasticity	2821.43***	0.0000
Born and Breitung serial correlation LM(k)-test	2.02**	0.044

Note: ** $p < 0.05$, *** $p < 0.01$.

Additionally, the Pesaran (2007) CIPS and CADF unit root tests show that the variables exhibit both level and first difference stationarity. Specifically, BFAITI, GDPG, and REG are stationary at the level in the CIPS test, while BFAITI and BRC are stationary at the level in the CADF test. However, all variables are stationary in both tests at the first difference, as demonstrated in Table 4.

4.2. Baseline estimations

Table 5 presents both the baseline and moderation results. According to column (1) of Table 4, BFAITI demonstrates a significant positive impact on BROA at the 5 % level. This suggests that a 1 % increase in banking and finance AI technology innovation leads to a 0.000899 increase in banks' financial performance. AI-driven predictive analytics and risk modeling tools allow banks to better assess and manage credit, market, and operational risks. By analyzing vast amounts of data and identifying patterns and correlations, AI systems can enhance credit underwriting, detect fraudulent activities, and improve portfolio management (Hassan, Aziz, & Andriansyah, 2023). This helps mitigate risks and reduces the likelihood of loan defaults or financial losses, ultimately contributing to higher returns on assets. AI-powered chatbots, virtual assistants, and recommendation engines enable banks to deliver personalized services, anticipate customer needs, and provide timely assistance (Patil, 2023). By leveraging customer data and behavioral insights, banks can offer tailored product recommendations, enhance

cross-selling opportunities, and improve overall customer satisfaction and loyalty (Rane, 2023). This can lead to increased customer retention, expanded market share, and ultimately higher profitability. Investing in AI technology innovation allows banks to stay ahead of the curve and gain a competitive advantage in the rapidly evolving financial landscape (Hu & Wu, 2023). By developing proprietary AI algorithms, partnering with fintech startups, or adopting cutting-edge AI solutions, banks can differentiate themselves, attract new customers, and capture new revenue streams (Moro-Visconti, 2024). This innovation-driven growth can drive higher returns on assets over time. This outcome conforms to the findings of Shiyab et al. (2023), showing that AI disclosures positively impact financial performance metrics like ROA and ROE and reduce total expenses, thus supporting the prevailing notion that AI enhances revenue and curtails costs. Their integrated analysis reveals a consistent

Table 5

Baseline regression and moderating effect results.

Variables	(1)	(2)
	BROA	BROA
BFAITI	0.000899** (2.36)	−0.00216** (−2.21)
BNL	−0.0703*** (−6.84)	−0.0720*** (−7.00)
BRC	0.0146** (2.17)	0.0163** (2.40)
GDPG	0.0327*** (4.39)	0.0333*** (4.50)
REG	−0.208*** (−5.51)	−0.256*** (−6.70)
BFAITI*GDPG		0.00181** (2.35)
Constant	0.729*** (5.86)	0.769*** (6.15)
Observations	220	220
Wald chi ²	105.25***	127.06***

Note: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$; z statistics in parentheses.**Table 4**

CSD, CIPS, and CADF tests.

Variables	CSD test	CIPS		CADF	
		Level	1st Difference	Level	1st Difference
BROA	25.568245***	−2.121	−3.457***	−1.803	−2.364**
BFAITI	25.950219***	−4.053***	−4.320***	−3.006***	−3.505***
BNL	38.3998***	−1.957	−2.673***	−1.716	−2.207**
BRC	41.136116***	−1.991	−2.590**	−2.356***	−3.005***
GDPG	27.991821***	−2.373**	−3.461***	−1.982	−2.114*
REG	14.594454***	−1.917	−3.085***	−1.405	−2.286**
ICTD	41.253054***	−2.435**	−3.369***	−1.776	−2.144*

Note: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

trend toward AI utilization among Jordanian banks, underscoring its transformative potential.

Conversely, BNPL shows a negative effect on BROA at the 1 % significance level, indicating that banks' non-performing loans reduce financial performance by -0.0703 . Non-performing loans indicate that a portion of the bank's loan portfolio is not generating income, as borrowers are unable to repay their loans. This suggests a deterioration in asset quality and an increase in credit risk for the bank (Yurtadur, Celiktas, & Celiktas, 2019). As non-performing loans accumulate, the bank may need to allocate more resources to provisions for loan losses, reducing its net income and profitability (Thornton & Di Tommaso, 2021). This, in turn, lowers the return on assets, as assets are not generating sufficient returns to cover the associated risks and costs. Non-performing loans can erode a bank's capital base, as they may need to set aside more capital to cover potential losses from defaults (Rahaman & Sur, 2021). This reduces the bank's capital adequacy ratio, which is a measure of its ability to absorb losses and maintain financial stability (Ozili, 2020). A lower capital adequacy ratio may lead to regulatory penalties or constraints on the bank's lending activities, constraining its ability to generate returns on its assets (Buchak, Matvos, Piskorski, & Seru, 2024). High levels of non-performing loans can signal financial distress and poor management practices to investors and stakeholders, leading to a loss of confidence in the bank's ability to generate profits and maintain solvency. This can result in a decline in the bank's stock price and market valuation, further impacting its financial performance. This finding aligns seamlessly with previous literature, such as (Duong, Tran, Nguyen, & Pham, 2023), underscoring the adverse impact of non-performing loans on financial performance, as evidenced by the 0.075 percentage point reduction in ROA per percentage point increase in NPLs in Vietnamese banks. This study's findings further resonate with the research by Takahashi and Vasconcelos (2024), which also indicates that NPLs diminish financial performance in the Brazilian banking sector. The agreement between these studies highlights the critical need for effective NPL management to enhance bank efficiency and profitability.

Moreover, the findings indicate that BRC positively influences BROA, with banks' regulatory capital enhancing financial performance by 0.0146 at the 5 % significance level. Regulatory capital represents the buffer or cushion that banks maintain to absorb potential losses from unexpected events, such as loan defaults or economic downturns. A higher level of regulatory capital indicates that banks are better equipped to withstand adverse shocks without jeopardizing their solvency (Saadaoui & Mokdadi, 2023). By maintaining higher regulatory capital levels, banks can reduce the probability of default and credit losses associated with their loan portfolios. This, in turn, leads to lower provisioning for loan losses and higher net income, thereby enhancing their return on assets. This finding is consistent with prior research conducted by De Bandt et al. (2018), who also identified a positive relationship between regulatory capital and financial performance in France. Their findings suggest that voluntary capital, not constrained by regulatory mandates, is the sole driver of enhanced performance, while regulatory capital has an insignificant impact on profitability. This integrated analysis highlights the importance of understanding the different roles that voluntary and regulatory capital play in shaping bank performance, highlighting both the agreement with prior research and the novel insight into the unique contribution of voluntary capital.

Similarly, economic growth is associated with increased banks' financial performance by 0.0327 at the 1 % significance level, as depicted in column (1) of Table 5. Economic growth stimulates demand for loans as businesses seek financing for expansion, investment, and working capital needs (Wang, Yang, Tang, & Wei, 2024). Similarly, individuals may borrow more for consumption purposes, such as purchasing homes or vehicles. As loan demand increases, banks can generate higher interest income from lending activities, thereby improving their financial performance (Kar & Swain, 2014). Economic growth creates a favorable environment for investment in various

sectors of the economy (Reis Mourão & Popescu, 2023). Banks can capitalize on these opportunities by providing financing for infrastructure projects, business expansions, and innovation initiatives. By participating in economic development activities, banks can generate additional revenue streams and diversify their income sources. Rising economic prosperity enhances consumer confidence and increases discretionary spending (Jonkman, Boukes, & Vliegthart, 2020). This can lead to higher demand for retail banking products and services, such as mortgages, credit cards, and wealth management services. Banks can capitalize on this trend by expanding their customer base and cross-selling additional products, thereby boosting their financial performance. This discovery aligns with the research inquiry conducted by reinforcing the notion that economic growth enhances banks' financial performance. It particularly highlights the positive association between interest margins, return on assets, and GDP. However, the outcome is contrary to the results of Robin, Salim, and Bloch (2018) which highlight a significant negative relationship between GDP growth rate and net interest margin as a measure of financial performance, indicating a counterintuitive dynamic where economic expansion might constrain interest margins. This contradiction suggests that while economic growth benefits overall financial health, it may exert pressure on banks' profitability margins through competitive lending practices or regulatory effects. Furthermore, government regulations have a negative impact on BROA at the 1 % significance level, suggesting that such regulations decrease banks' financial performance by 0.208 %. Government regulations, such as interest rate caps or restrictions on lending practices, can influence banks' interest income and net interest margins (Ferrari et al., 2018). If regulatory measures limit banks' ability to adjust interest rates or engage in profitable lending activities, it can depress overall profitability and reduce BROA. Regulatory restrictions may limit the range of products and services that banks can offer, impose constraints on risk-taking behavior, or restrict entry into new markets (Haque, 2019). These limitations can hinder banks' ability to generate profits and optimize their asset utilization, leading to a decrease in BROA. This finding corresponds to the investigation conducted by Batae et al. (2021), which unveiled that an enhancement in the efficacy of a bank's corporate governance framework was observed to detrimentally impact the financial outcomes of European banks, underscoring a pivotal external element influencing these dynamics.

Introducing the interactive term, BRAITI*GDPG, into the main equation (see Eq. (7)) in column (2) of Table 5, reveals a positive and significant coefficient for the interaction between BRAITI and GDPG on banks' financial performance at the 1 % level. However, the level coefficients of BRAITI and GDPG are negative and positive, respectively. This implies that the application of banking and finance AI technology innovation should be coupled with economic growth rather than being implemented alone. The negative coefficient for BRAITI could be due to factors such as initial investment costs, implementation challenges, or disruptions to existing business processes associated with AI adoption. Conversely, the positive coefficient for GDPG aligns with economic theory, as stronger economic growth typically leads to increased demand for financial services, higher investment opportunities, and improved credit quality, all of which can boost banks' profitability (Salike & Ao, 2018). The positive and significant coefficient for the interaction term BRAITI*GDPG indicates that the combined effect of banking and finance AI technology innovation (BRAITI) and economic growth on banks' financial performance is greater than the sum of their individual effects. This suggests that when AI technology innovation is coupled with favorable economic growth conditions, the positive impact on banks' financial performance is amplified.

4.3. Mediation effect

Table 6 illustrates the outcomes of the intermediary role of ICT development in the nexus between BFAITI and banks' financial performance. In column (1), the primary regression results are displayed.

Table 6
Mediating effect results.

Variables	(1)	(2)	(3)
	BROA	ICTD	BROA
BFAITI	0.000899** (2.36)	0.0443* (1.76)	0.000713* (1.62)
ICTD			0.00123 (1.23)
Constant	0.729*** (5.86)	127.8*** (21.33)	0.601*** (2.99)
Control variables	YES	YES	YES
Observations	220	220	220
Wald chi ²	105.25***	333.27***	114.71***

Note: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$; z statistics in parentheses.

Upon examination of column (2), derived from Eqs. (9), it is evident that BFAITI positively influences ICTD at a significance level of 10 %, indicating that BFAITI contributes to ICT development. This can be attributed to the need for advanced ICT systems to support the integration and utilization of AI technologies in banking operations, such as data analytics, automation, and customer interaction platforms (Naimi-Sadigh, Asgari, & Rabiei, 2022).

Moving on to column (3), we observe a positive coefficient for ICTD at the 10 % significance level, suggesting substantial direct effects of ICTD on BROA. Moreover, the positive and significant coefficients for BFAITI indicate that ICTD completely mediates the enhancement of banks' financial performance. This can be explained by the role of ICT in improving operational efficiency, reducing costs, enhancing customer experience, and facilitating innovation in financial products and services (Abdurrahman, Gustomo, & Prasetyo, 2024). For example, advanced ICT systems enable banks to streamline processes, deliver personalized services, and adapt quickly to changing market conditions, ultimately leading to improved profitability and financial outcomes. The significant coefficients for BROA suggest that ICTD completely mediates the enhancement of banks' financial performance. This implies that the positive effects of BFAITI on financial performance are fully transmitted through improvements in ICT infrastructure. As banks invest in and leverage AI technologies to innovate and optimize their operations, the resulting improvements in ICT infrastructure enable them to realize the full potential of AI-driven initiatives, leading to enhanced financial performance (Thowfeek, Samsudeen, & Sanjeetha, 2020). This mediation effect highlights the critical role of ICT development as a channel for translating AI innovation into tangible financial benefits for banks.

4.4. Lagged effects

Banks' financial performance is a dynamic and ongoing process that unfolds over time. When assessing the influence of AI technology innovation in banking and finance on shifts in bank financial performance, it is crucial to account for its lagged effects. The impact of AI innovations introduced in the banking and finance sector during a particular year may not be immediately apparent; instead, it might manifest several years later. Selecting the initial lag as the timeframe for the lagged effects of AI technology innovation in banking and finance to manifest could potentially lead to inaccuracies in estimating the effects of such innovations on changes in bank financial performance. Recognizing the issue of lagged effects associated with AI technology innovation in banking and finance, this study incorporates the strength of AI technology innovation in banking and finance from the first, second, third, fourth, and fifth lag periods as independent variables into Eq. (7), yielding Eq. (11). The regression outcomes are presented in Table 7. The results presented in Table 7 illustrate the varying impact of banking and finance AI technology innovations on banks' financial performance across different lag periods. Eq. (11) estimation outcomes reveal that the coefficients in the table signify the influence of BFAITI on banks' financial performance, considering lag periods ranging from 1 to 5. In the initial four periods, an upsurge in banking and finance AI technology innovation corresponds with an improvement in banks' financial performance, albeit with diminishing effect sizes and significance levels over time. This suggests that the immediate response to banking and finance AI technology innovation is an enhancement in banks' financial performance. However, from lag 5 onward, there is a transition to a negative relationship, indicating that prolonged exposure to the costs of banking and finance AI technology innovation or the erosion of initial benefits may result in a decline in banks' financial performance.

Similarly, these coefficients also reflect the impacts of banking and finance AI technology innovation on banks' financial performances, with corresponding increases observed over four lag periods. Although statistically significant at lags 1 and 2, this increase, ranging from a 0.000898 (lag 1) to a 0.000901 (lag 2) rise in banks' financial performances, is potentially modest. The diminishing effect sizes and significance levels observed over time indicate potential diminishing returns to AI technology innovation in banking and finance. Initially, the adoption of AI technologies may yield substantial improvements in financial performance. However, as AI technologies mature and become more widely adopted across the industry, the competitive advantage gained from early adoption may diminish, resulting in a leveling off of financial performance improvements. The transition to a negative relationship between AI technology innovation and financial performance observed after a certain lag period suggests that banks may reach a point where

Table 7
Lagged effects results.

Variables	(1)	(2)	(3)	(4)	(5)
	BROA	BROA	BROA	BROA	BROA
BFAITI-1	0.000898** (2.50)				
BFAITI-2		0.000901* (1.94)			
BFAITI-3			0.000839 (1.35)		
BFAITI-4				0.000488 (0.61)	
BFAITI-5					-0.000342 (-0.40)
Constant	0.648*** (39.72)	0.646*** (39.59)	0.644*** (39.33)	0.645*** (39.08)	0.648*** (39.29)
Control variables	YES	YES	YES	YES	YES
Observations	219	218	217	216	215
Wald chi ²	6.23**	3.77*	1.83	0.37	0.16

Note: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$; z statistics in parentheses.

the costs associated with sustaining or scaling AI initiatives outweigh the benefits. This could be due to factors such as escalating implementation costs, ongoing maintenance expenses, or diminishing competitive advantages as AI becomes more ubiquitous in the industry.

4.5. Robustness tests

Table 8 below displays the results of the robustness tests conducted by altering the methodology and substituting the dependent variable. In this study, the two-step system generalized method of moments (GMM) technique was employed to evaluate the validity of the baseline model. This method is particularly suitable for datasets exhibiting cross-sectional dependence, heterogeneity, and individual time effects, and it effectively addresses endogeneity issues (Baffour Gyau, Li, & Adu, 2024). The results obtained using this method are consistently reliable. In column (1), there is a sustained positive association between banking and finance AI technology innovations and banks' financial performance, indicating that a 1 % increase in banking and finance AI technology innovation corresponds to a similar increase in banks' financial performance. Additionally, to further corroborate the baseline findings, the dependent variable was replaced. The banks' financial performance variable was substituted with banks' return on equity (BROE), as depicted in column (2). The results in column (2) demonstrate that, even with this modification in measurement methodology, the coefficient for BFAITI remains significantly positive at 10 %, thus ensuring the continued validity of the results. The analysis presented above indicates that the empirical findings from the main estimations remain consistent and reliable.

5. Conclusion, policy recommendations, limitations, and directions for future research

This study delves into the complex relationship between banking and finance AI technology innovation and the financial performance of banks. It analyzes panel data from 20 economies, employing the FGLS method for primary estimations and the two-step system GMM technique for conducting robustness checks. This study seeks to explore the following research questions: Does AI technology innovation enhance the financial performance of banks? What mechanisms drive this relationship? Additionally, does AI technology innovation impact banks' financial performance across different lag periods? The findings show that banking and finance AI technology innovation has a positive impact on banks' return on assets, indicating that increased innovation in banking and finance AI technology enhances banks financial performance. Conversely, non-performing loans have a negative effect on BROA, emphasizing the detrimental impact of asset quality on banks'

profitability. Additionally, regulatory capital and economic growth show positive associations with BROA, suggesting that stronger regulatory capital and a conducive economic environment contribute to improved financial performance. However, government regulations exhibit a negative impact on BROA, highlighting the challenges imposed by regulatory frameworks on banks' profitability. Moreover, the interaction between BFAITI and economic growth highlights the importance of coupling AI technology innovation with economic growth to maximize its positive impact on banks' financial performance. Furthermore, the mediation analysis illustrates that ICT development plays a crucial role in mediating the relationship between BFAITI and banks' financial performance, indicating that the enhancement of ICT infrastructure facilitates the translation of AI technology innovation into improved financial outcomes for banks. Considering the lagged effects of AI technology innovation, the study reveals a dynamic relationship between BFAITI and banks' financial performance over time. While an initial surge in innovation leads to an immediate improvement in financial performance, prolonged exposure to innovation costs or diminishing benefits may result in a decline in performance over time. Finally, the robustness analysis confirms the consistency and reliability of the baseline findings, reinforcing the positive association between BFAITI and banks' financial performance across different methodologies and measurement variables. This study highlights the significant role of banking and finance AI technology innovation in boosting banks' financial performance, while also emphasizing the necessity of integrating contextual factors such as economic growth, regulatory frameworks, and ICT development to fully leverage AI's benefits. Notably, it highlights a significant gap in the literature regarding the complex interactions between these factors and AI adoption, calling for deeper exploration to optimize strategic implementation in the banking sector.

Drawing from the findings outlined above, several policy implications can be suggested to harness the potential benefits of banking and finance AI technology innovation while addressing associated challenges. First, there should be the promotion of AI innovation ecosystem. Governments and regulatory bodies in the 20 countries should incentivize and support the development of an AI innovation ecosystem within the banking and finance sector. This includes funding research and development initiatives, providing tax incentives for AI investment, and fostering collaboration between financial institutions, technology firms, and academic institutions. By doing so, policymakers can stimulate innovation, enhance the competitiveness of the banking sector, and create an environment conducive to AI-driven growth. Second, regulators need to update and adapt regulatory frameworks to accommodate the rapid advancements in AI technology while ensuring consumer protection, data privacy, and financial stability. This involves establishing clear guidelines for AI implementation in banking, conducting regular audits of AI systems, and promoting transparency and explainable AI-driven decision-making processes. Such measures will help build trust in AI systems, safeguard against potential risks, and ensure that AI adoption aligns with regulatory standards. Third, policymakers should prioritize investment in ICT infrastructure to support the adoption and implementation of AI technologies in the banking sector. This includes expanding broadband access, enhancing cybersecurity measures, and promoting digital literacy among the population. Strengthening ICT infrastructure is crucial for maximizing the benefits of AI technology and ensuring that all stakeholders can effectively leverage AI capabilities. Fourth, financial institutions should focus on improving risk management practices and asset quality to mitigate the negative impact of non-performing loans on financial performance. This may involve implementing stricter lending standards, enhancing credit risk assessment models using AI, and diversifying loan portfolios to reduce concentration risk. By adopting robust risk management strategies, banks can enhance their resilience and stability in the face of economic fluctuations. Fifth, governments should implement policies aimed at fostering economic growth and promoting innovation synergy. This includes promoting entrepreneurship, investing in education and skills

Table 8
Robustness tests results.

Variables	(1)	(2)
	2-step system GMM	BROE
<i>L.BROA</i>	0.120* (1.85)	
<i>BFAITI</i>	0.000411*** (3.68)	0.00273* (1.90)
Constant	0.663*** (3.40)	12.58*** (6.20)
Control variables	YES	YES
Observations	200	220
F statistics/ Wald chi ²	295.6***	58.29***
Instruments	18	
Groups	20	
AR (1)	−1.79*	
AR (2)	1.19	
Sargan test	116.37***	
Hansen test	12.94	

Note: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$; z statistics in parentheses.

development in STEM fields, and facilitating collaboration between the public and private sectors to leverage AI technology for economic development. By aligning AI innovation with economic growth strategies, countries can create a sustainable environment for AI adoption and utilization. Sixth, financial institutions should adopt long-term strategic planning approaches that account for the dynamic nature of AI technology innovation and its potential lagged effects on financial performance. This involves conducting regular assessments of AI implementation strategies, monitoring industry trends, and adjusting business models accordingly. Strategic planning will enable banks to stay ahead of technological advancements and continuously optimize their AI-driven initiatives. Seventh, policymakers and regulators should establish mechanisms for continuous monitoring and evaluation of AI adoption in the banking sector. This ensures compliance with regulations, assesses potential risks, and measures the effectiveness of AI-driven initiatives in improving financial performance and customer outcomes. Continuous monitoring will help maintain the integrity and reliability of AI systems while fostering innovation. Lastly, international collaboration and knowledge-sharing initiatives should be encouraged to facilitate the exchange of best practices, lessons learned, and regulatory experiences related to AI adoption in the banking and finance sector. Promoting global standards and frameworks for responsible AI governance will foster innovation, enhance competitiveness, and ensure that AI technology contributes to sustainable development goals. By implementing these policy recommendations, stakeholders can effectively leverage banking and finance AI technology innovation to drive sustainable economic growth, enhance financial stability, and improve the overall well-being of society. This study not only informs policy-making in the 20 countries but also provides actionable insights for practitioners, researchers, and global efforts toward achieving sustainable development goals.

The research has few limitations despite the valuable empirical discoveries. Firstly, the investigation is confined to the period from 2010 to 2020, primarily due to constraints related to the dataset. Future research endeavors could enhance comprehensiveness by extending the dataset to incorporate more recent years. Secondly, this study focuses on the moderating influence of economic growth; however, forthcoming research should contemplate additional factors that might play a moderating role. Thirdly, expanding the scope to encompass a more extensive array of countries, provided there is available data, could provide a more detailed understanding of the connections. Lastly, upcoming research endeavors could derive benefits from undertaking heterogeneous analyses, taking into account factors like the income levels or developmental statuses of countries.

Data availability

Data will be made available on request.

References

- Abdurrahman, A., Gustomo, A., & Prasetyo, E. A. (2024). Impact of dynamic capabilities on digital transformation and innovation to improve banking performance: A TOE framework study. *Journal of Open Innovation: Technology, Market, and Complexity*, 100215.
- Abrokwa-Larbi, K., & Awuku-Larbi, Y. (2024). The impact of artificial intelligence in marketing on the performance of business organizations: Evidence from SMEs in an emerging economy. *Journal of Entrepreneurship in Emerging Economies*, 16(4), 1090–1117.
- Adu, D. A., Abedin, M. Z., Saa, V. Y., & Boateng, F. (2024). Bank sustainability, climate change initiatives and financial performance: The role of corporate governance. *International Review of Financial Analysis*, 103438.
- Aguegbah, E. S., Agu, C. V., & Nnetu-Okolieuwa, V. I. (2023). ICT adoption, bank performance & development in sub-Saharan Africa: A dynamic panel analysis. *Information Technology for Development*, 29(2–3), 406–422.
- Alam, M. S., Rabbani, M. R., Tausif, M. R., & Abey, J. (2021). Banks' performance and economic growth in India: A panel cointegration analysis. *Economics*, 9(1), 38.
- Al-Busaidi, K. A., & Al-Muharrami, S. (2021). Beyond profitability: ICT investments and financial institutions performance measures in developing economies. *Journal of Enterprise Information Management*, 34(3), 900–921.
- Anginer, D., Demircug-Kunt, A., & Zhu, M. (2014). How does competition affect bank systemic risk? *Journal of Financial Intermediation*, 23(1), 1–26.
- Appiah, M., Ashraf, S., Tiwari, A. K., Gyamfi, B. A., & Onifade, S. T. (2023). Does financialization enhance renewable energy development in sub-Saharan African countries? *Energy Economics*, 125, Article 106898.
- Appiah, M., Karim, S., Naeem, M. A., & Lucey, B. M. (2022). Do institutional affiliation affect the renewable energy-growth nexus in the sub-Saharan Africa: Evidence from a multi-quantitative approach. *Renewable Energy*, 191, 785–795.
- Asongu, S. A., & Biekpe, N. (2018). ICT, information asymmetry and market power in African banking industry. *Research in International Business and Finance*, 44, 518–531.
- Athanasoglou, P. P., Brissimis, S. N., & Delis, M. D. (2008). Bank-specific, industry-specific and macroeconomic determinants of bank profitability. *Journal of international financial Markets, Institutions and Money*, 18(2), 121–136.
- Baffour Gyau, E., Li, Y., & Adu, D. (2024). Investigating the impact of ICT on transport-based CO2 emissions: empirical evidence from a quantile cointegration regression analysis. *Environmental Science and Pollution Research*, 31(3), 4606–4629.
- Bai, J., Choi, S. H., & Liao, Y. (2021). Feasible generalized least squares for panel data with cross-sectional and serial correlations. *Empirical Economics*, 60, 309–326.
- Banna, H., & Alam, M. R. (2021). Impact of digital financial inclusion on ASEAN banking stability: Implications for the post-Covid-19 era. *Studies in Economics and Finance*, 38(2), 504–523.
- Barney, J. (1991). Firm resources and sustained competitive advantage. *Journal of Management*, 17(1), 99–120.
- Bătae, O. M., Dragomir, V. D., & Felegă, L. (2021). The relationship between environmental, social, and financial performance in the banking sector: A European study. *Journal of Cleaner Production*, 290, Article 125791.
- Born, B., & Breitung, J. (2016). Testing for serial correlation in fixed-effects panel data models. *Econometric Reviews*, 35(7), 1290–1316.
- Breitung, J., & Das, S. (2005). Panel unit root tests under cross-sectional dependence. *Statistica Neerlandica*, 59(4), 414–433.
- Buchak, G., Matvos, G., Piskorski, T., & Seru, A. (2024). Beyond the balance sheet model of banking: Implications for bank regulation and monetary policy. *Journal of Political Economy*, 132(2), 616–693.
- Center for Security and Emerging Technology. (2023). Annual granted patents related to artificial intelligence, by industry. Our world in data. Retrieved February 22, 2024 from <https://ourworldindata.org/artificial-intelligence>.
- Center for Security and Emerging Technology (2023). Country Activity Tracker (CAT): Artificial Intelligence "Banking and Finance". Our World in Data. Retrieved February 21, 2024 from <https://cat.eto.tech/?expanded=Summary+metrics&dataset=Patent&patentField=Banking+and+Finance>.
- Chhaidar, A., Abdelhedi, M., & Abdelkafi, I. (2023). The effect of financial technology investment level on European banks' profitability. *Journal of the Knowledge Economy*, 14(3), 2959–2981.
- De Bandt, O., Camara, B., Maitre, A., & Pessarossi, P. (2018). Optimal capital, regulatory requirements and bank performance in times of crisis: Evidence from France. *Journal of Financial Stability*, 39, 175–186.
- Del Gaudio, B. L., Porzio, C., Sampagnaro, G., & Verdoliva, V. (2021). How do mobile, internet and ICT diffusion affect the banking industry? An empirical analysis. *European Management Journal*, 39(3), 327–332.
- Duong, K. D., Tran, P. M. D., Nguyen, P. Y. N., & Pham, H. (2023). How do funding diversity and non-performing loans affect bank performance in different economic cycles? *Cogent Business & Management*, 10(2), 2215076.
- Dutta, S., Pramanik, H. S., Datta, S., & Kirtania, M. (2022). *Imperatives, Trends and Dynamics of Digital Transformation as Banks Adopt Technology and Intelligent Systems. In Intelligent Systems in Digital Transformation: Theory and Applications* (pp. 323–348). Cham: Springer International Publishing.
- Engvall, N. (2024). *The Influence of Institutional Factors on AI adoption in EU banking cybersecurity: A narrative literature review*.
- Eni, L. N., Chaudhary, K., Raparthi, M., & Reddy, R. (2023). Evaluating the role of artificial intelligence and big data analytics in Indian Bank marketing. *Tuijin Jishu/ Journal of Propulsion Technology*, 44.
- Ferrari, A., Masetti, O., & Ren, J. (2018). Interest rate caps: The theory and the practice. In , 8398. *World Bank Policy research working paper*.
- Fomby, T. B., Johnson, S. R., Hill, R. C., Fomby, T. B., Johnson, S. R., & Hill, R. C. (1984). Feasible generalized least squares estimation. *Advanced econometric methods*, 147–169.
- Galaritis, E., Kosmidou, K., Kousenidis, D., Lazaridou, E., & Papapanagiotou, T. (2021). Measuring the effects of M&As on Eurozone bank efficiency: An innovative approach on concentration and credibility impacts. *Annals of Operations Research*, 306, 343–368.
- Ghaemi, S., & Hosseini, M. H. (2023). Modeling construction freight using OLS, robust inference and FGLS. *Iranian Journal of Science and Technology, Transactions of Civil Engineering*, 1–18.
- Gyamfi, B. A., Ozturk, I., Bein, M. A., & Bekun, F. V. (2021). An investigation into the anthropogenic effect of biomass energy utilization and economic sustainability on environmental degradation in E7 economies. *Biofuels, Bioproducts and Biorefining*, 15(3), 840–851.
- Haibo, C., & Manu, E. K. (2022). The impact of banks' financial performance on environmental performance in Africa. *Environmental Science and Pollution Research*, 29(32), 49214–49233.
- Hakimi, A., Hamdi, H., & Khemiri, M. A. (2023). Banking in the MENA region: The proactive role of financial and economic freedom. *Journal of Policy Modeling*, 45(5), 1058–1076.
- Haque, F. (2019). Ownership, regulation and bank risk-taking: Evidence from the Middle East and North Africa (MENA) region. *Corporate Governance: The international journal of business in society*, 19(1), 23–43.

- Hassan, M., Aziz, L. A.-R., & Andriansyah, Y. (2023). The role artificial intelligence in modern banking: An exploration of AI-driven approaches for enhanced fraud prevention, risk management, and regulatory compliance. *Reviews of Contemporary Business Analytics*, 6(1), 110–132.
- Hu, B., & Wu, Y. (2023). AI-based compliance automation in commercial bank: How the silicon valley bank provided a cautionary tale for future integration. *International Research in Economics and Finance*, 7(1), 13.
- Imran, M., Khan, M. K., Alam, S., Wahab, S., Tufail, M., & Jijian, Z. (2024). The implications of the ecological footprint and renewable energy usage on the financial stability of south Asian countries. *Financial Innovation*, 10(1), 102.
- International Monetary Fund. (2023). *Return on assets. Our World in Data*. Retrieved February 21, 2024 from <https://ourworldindata.org/grapher/regulation-financial-markets?time=2010..2020&country=AUS~BRA~CAN~CHN~DNK~FRA~DEU~IND~ISR~ITA~KOR~MYS~MEX~RUS~SGP~ESP~SWE~GBR~USA>.
- Jonkman, J., Boukes, M., & Vliegthart, R. (2020). When do media matter most? A study on the relationship between negative economic news and consumer confidence across the twenty-eight EU states. *The International Journal of Press/Politics*, 25(1), 76–95.
- Kar, A. K., & Swain, R. B. (2014). Interest rates and financial performance of microfinance institutions: Recent global evidence. *The European Journal of Development Research*, 26, 87–106.
- Kaya, O., Schildbach, J., AG, D. B., & Schneider, S. (2019). Artificial intelligence in banking. *Artificial*. [https://www.dbresearch.com/PROD/RPS_EN-PROD/P ROD0000000000495172/Artificial intelligence in banking: A lever for pr.pdf?undefined&reaload=8ZyKXL9MauITRfzILKybIDVYrVM4H8AEQdpXenTZjzm2AJyaFlhr4dktmULaWzu](https://www.dbresearch.com/PROD/RPS_EN-PROD/P ROD0000000000495172/Artificial%20intelligence%20in%20banking%20A%20lever%20for%20pr.pdf?undefined&reaload=8ZyKXL9MauITRfzILKybIDVYrVM4H8AEQdpXenTZjzm2AJyaFlhr4dktmULaWzu).
- Königstorfer, F., & Thalmann, S. (2020). Applications of artificial intelligence in commercial banks—a research agenda for behavioral finance. *Journal of Behavioral and Experimental Finance*, 27, Article 100352.
- Kreps, S., & Rao, A. (2023). *AI and the Regulatory Challenge: A New Framework Using the Seto Loop*. Available at SSRN.
- Ledhem, M. A., & Mekidiche, M. (2020). Economic growth and financial performance of Islamic banks: A CAMELS approach. *Islamic Economic Studies*, 28(1), 47–62.
- Li, Y., & Zhang, Y. (2023). What is the role of green ICT innovation in lowering carbon emissions in China? A provincial-level analysis. *Energy Economics*, 127, Article 107112.
- Lütkepohl, H., & Burda, M. M. (1997). Modified Wald tests under nonregular conditions. *Journal of Econometrics*, 78(2), 315–332.
- Madhura, K., Panakaje, N., Kambali, U., & Parvin, S. R. (2024). Unveiling the Progress of artificial intelligence in the banking landscape of India. In , Vol. 2. *The AI revolution: Driving business innovation and research* (pp. 497–510). Springer.
- Moro-Visconti, R. (2024). The Valuation of Artificial Intelligence-Driven Startups. In *Artificial Intelligence Valuation: The Impact on Automation, BioTech, ChatBots, FinTech, B2B2C, and Other Industries* (pp. 293–344). Springer.
- Naimi-Sadigh, A., Asgari, T., & Rabiei, M. (2022). Digital transformation in the value chain disruption of banking services. *Journal of the Knowledge Economy*, 13(2), 1212–1242.
- Nasreen, S., Gulzar, M., Afzal, M., & Farooq, M. U. (2024). The role of corruption, transparency, and regulations on Asian banks' performance: an empirical analysis. *Journal of the Knowledge Economy*, 15(2), 6475–6506.
- Nsanyan Sandow, J., Duodu, E., & Otegh-Abayie, E. F. (2021). Regulatory capital requirements and bank performance in Ghana: evidence from panel corrected standard error. *Cogent Economics & Finance*, 9(1), 2003503.
- Ozili, P. K. (2020). Non-performing loans in European systemic and non-systemic banks. *Journal of Financial Economic Policy*, 12(3), 409–424.
- Papadopoulos, A. (2024). Some notes on the asymmetry of the regression error. *Journal of Productivity Analysis*, 61(1), 37–42.
- Partovi, E., & Matousek, R. (2019). Bank efficiency and non-performing loans: Evidence from Turkey. *Research in International Business and Finance*, 48, 287–309.
- Patil, A. (2023). *AI-powered banking: From customer service to risk management*.
- Pesaran, M. H. (2007). A simple panel unit root test in the presence of cross-section dependence. *Journal of Applied Econometrics*, 22(2), 265–312.
- Pesaran, M. H. (2021). General diagnostic tests for cross-sectional dependence in panels. *Empirical Economics*, 60(1), 13–50.
- Pesaran, M. H., & Yamagata, T. (2008). Testing slope homogeneity in large panels. *Journal of Econometrics*, 142(1), 50–93.
- Polak, P., Nelischer, C., Guo, H., & Robertson, D. C. (2020). Intelligent finance and treasury management: what we can expect. *AI & Society*, 35(3), 715–726.
- Rahaman, S. M., & Sur, D. (2021). Identifying Key Drivers of Non-Performing Assets in Indian Public Sector Banks: A Panel Data Analysis. *IIM Kozhikode Society & Management Review*, 0(0). <https://doi.org/10.1177/22779752211000146>
- Rahman, M., Ming, T. H., Baigh, T. A., & Sarker, M. (2023). Adoption of artificial intelligence in banking services: An empirical analysis. *International Journal of Emerging Markets*, 18(10), 4270–4300.
- Rahman, M. M. (2023). The effect of business intelligence on bank operational efficiency and perceptions of profitability. *FinTech*, 2(1), 99–119.
- Rane, N. Enhancing customer loyalty through Artificial Intelligence (AI), Internet of Things (IoT), and Big Data technologies: improving customer satisfaction, engagement, relationship, and experience. Internet of Things (IoT), and Big Data Technologies: Improving Customer Satisfaction, Engagement, Relationship, and Experience. <https://doi.org/10.2139/ssrn.4616051>.
- Reis Mourão, P. J., & Popescu, I. A. (2023). Investment, growth and competitiveness: The multiplier-accelerator in the 21st century. *Journal of Competitiveness*, 15(3).
- Ris, K., Stankovic, Z., & Avramovic, Z. (2020). Implications of implementation of artificial intelligence in the banking business with correlation to the human factor. *Journal of Computer and Communications*, 8(11), 130.
- Robin, L., Salim, R., & Bloch, H. (2018). Financial performance of commercial banks in the post-reform era: Further evidence from Bangladesh. *Economic Analysis and Policy*, 58, 43–54.
- Rogers, E. M., Singhal, A., & Quinlan, M. M. (2014). Diffusion of innovations. In *An integrated approach to communication theory and research* (pp. 432–448). Routledge.
- Saadaoui, Z., & Mokdadi, S. (2023). Capital buffers, business models and the probability of bank distress: A dynamic panel investigation. *Journal of Financial Regulation and*, 31(5), 663–695.
- Salike, N., & Ao, B. (2018). Determinants of bank's profitability: Role of poor asset quality in Asia. *China Finance Review International*, 8(2), 216–231.
- Scott, R. (1995). *Institutions and organizations*. Thousand Oaks: SAGE Publications.
- Shao, J., Lou, Z., Wang, C., Mao, J., & Ye, A. (2022). The impact of artificial intelligence (AI) finance on financing constraints of non-SOE firms in emerging markets. *International Journal of Emerging Markets*, 17(4), 930–944.
- Shen, C. H., & Chang, Y. H. (2006). Do regulations affect banking performance? Government governance may matter. *Contemporary Economic Policy*, 24(1), 92–105.
- Shiyab, F. S., Alzoubi, A. B., Obidat, Q. M., & Alshurafat, H. (2023). The impact of artificial intelligence disclosure on financial performance. *International Journal of Financial Studies*, 11(3), 115.
- Siddika, A., & Haron, R. (2020). Capital regulation and ownership structure on bank risk. *Journal of Financial Regulation and Compliance*, 28(1), 39–56.
- Subburayan, B., Sankarkumar, A. V., Singh, R., & Mushi, H. M. (2024). Transforming of the financial landscape from 4.0 to 5.0: Exploring the integration of Blockchain, and artificial intelligence. *Applications of Block Chain Technology and Artificial Intelligence*, 137–161.
- Sullivan, Y., & Wamba, S. F. (2024). Artificial intelligence and adaptive response to market changes: A strategy to enhance firm performance and innovation. *Journal of Business Research*, 174, Article 114500.
- Takahashi, F. L., & Vasconcelos, M. R. (2024). Bank efficiency and undesirable output: An analysis of non-performing loans in the Brazilian banking sector. *Finance Research Letters*, 59, Article 104651.
- Thornton, J., & Di Tommaso, C. (2021). The effect of non-performing loans on credit expansion: Do capital and profitability matter? Evidence from European banks. *International journal of finance & economics*, 26(3), 4822–4839.
- Thowfeek, M. H., Samsudeen, S., & Sanjeetha, M. B. F. (2020). Drivers of artificial intelligence in banking service sectors. *Solid State Technology*, 63(5), 6400–6411.
- To, H. K., & Chan, K. W. (2024). Mean stationarity test in time series: A signal variance-based approach. *Bernoulli*, 30(2), 1231–1256.
- Tran, V. T., Lin, C.-T., & Nguyen, H. (2016). Liquidity creation, regulatory capital, and bank profitability. *International Review of Financial Analysis*, 48, 98–109.
- Truby, J., Brown, R., & Dahdal, A. (2020). Banking on AI: Mandating a proactive approach to AI regulation in the financial sector. *Law and Financial Markets Review*, 14(2), 110–120.
- Umamaheswari, S., & Valarmathi, A. (2023). Role of artificial intelligence in the banking sector. *Journal of Survey in Fisheries Sciences*, 10(4), 2841–2849.
- Wang, Y., Yang, S., Tang, W., & Wei, L. (2024). Government GDP targets and corporate capacity expansion—empirical evidence based on A-share listed companies. *International Review of Financial Analysis*, 91, Article 103048.
- World Bank. (2024). *World Development Indicators*. Retrieved February 21, 2024 from <https://databank.worldbank.org/source/world-development-indicators>.
- Yurttdur, M., Celiktas, E., & Celiktas, E. (2019). The place of non-performing loans in the Turkish banking sector. *Procedia Computer Science*, 158, 766–771.
- Zaki, M. (2019). Digital transformation: Harnessing digital technologies for the next generation of services. *Journal of Services Marketing*, 33(4), 429–435.
- Zeqiraj, V., Gurdiev, C., Sohag, K., & Hammoudeh, S. (2024). Economic uncertainty, public debt and non-performing loans in the Eurozone: Three systemic crises. *International Review of Financial Analysis*, 93, Article 103208.
- Zhao, Q., Tsai, P.-H., & Wang, J.-L. (2019). Improving financial service innovation strategies for enhancing china's banking industry competitive advantage during the fintech revolution: A Hybrid MCDM model. *Sustainability*, 11(5), 1419.